



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION



Academic Year: 2025–26

CRM EMPLOYEE MANAGEMENT SYSTEM

AN INTERNSHIP REPORT

*Submitted in partial fulfillment for the
Diploma Engineering in Computer Engineering
6th Semester, Silver Oak University, Ahmedabad*

Submitted by
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CERTIFICATE

This is to certify that the Internship report submitted along with the Internship entitled **CRM Employee Management System** has been carried out by **JOSHI YUG MUKEHBHAI**(Enrollment No.: 2302020400449) under my guidance in partial fulfillment for the Diploma Engineering in **Computer Engineering**, 6th Semester of Silver Oak University, Ahmedabad during the academic year 2025–26.

Mr. Nayan Meghwal
Internal Guide

Dr. Manish Singh
Head of Department C.E



Date: ____ / ____ / 2026

TO WHOM IT MAY CONCERN

This is to certify that **Joshi yug Maheshbhai**, a student with Enrollment No. 2302020400449, Semester 6th, Department of Computer Engineering, Aditya Silver Oak Institute of Technology has successfully completed his internship in the field of **Web Application Development – CRM Employee Management System** at **Vision Technolabs**, Ahmedabad under the guidance of the assigned Industry Mentor.

His internship activities include: Development of a full-stack CRM web application covering employee attendance management, task assignment and tracking, leave management, break time logging, holiday management, announcements, and admin dashboard with comprehensive reporting features.

During the period of his internship with us, he has been exposed to different processes and was found diligent, hardworking, and inquisitive. We wish him every success in his life and career.

DECLARATION

I hereby declare that the Internship report submitted along with the Internship entitled **CRM Employee Management System** submitted in partial fulfillment for the Diploma Engineering in **Computer Engineering** to Silver Oak University, Ahmedabad, is a bonafide record of original project work carried out by me at **Vision Technolabs** under the supervision of **Mr. Nayan Meghwal** and that no part of this report has been directly copied from any student's reports or taken from any other source, without providing due reference.

Name of Student

Joshi Yug Mukeshbhai

Sign of Student

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to all those who have helped and supported me throughout the completion of this internship and the preparation of this report.

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Finally, I am immensely thankful to my family and friends for their unwavering support, encouragement, and patience throughout this journey.

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ABSTRACT

The CRM Employee Management System is a comprehensive web-based application developed during the internship at Vision Technolabs, Ahmedabad. The primary objective of this project is to digitize and streamline the daily workforce management activities of an organization through an intuitive and user-friendly online portal. The system is designed to serve two primary user roles: the **Administrator** and the **Employee**.

The **Employee Module** enables staff members to log in securely, track their daily work hours through an active session timer, mark their attendance, log and track break durations, work on assigned tasks with dedicated task timers, apply for leaves and view their approval status, view upcoming holidays, team member birthdays, and receive company announcements. Employees can also manage their personal profile information.

The **Administrator Module** provides comprehensive oversight of all employee activities. The admin can add and manage employee profiles, view and monitor attendance records, approve or reject leave requests, create projects, assign tasks to employees, manually add work and break time logs, manage company holiday lists, post announcements, view detailed work-history reports, and manage their account password.

The system is built using modern web technologies with a robust MySQL database integration, ensuring real-time data processing, secure user authentication via session-based access control, and a scalable modular architecture.

LIST OF FIGURES

Figure No.	Figure Title	Page No.
Fig 5.1	Employee Login Page	30
Fig 5.2	Employee Dashboard – Active Session Timer	31
Fig 5.3	Employee Dashboard – Task Board (Pending Tasks)	32
Fig 5.4	Employee Dashboard – Completed Tasks	33
Fig 5.5	Employee Dashboard – Employee Break Activity	34
Fig 5.6	Employee Dashboard – Employee Status and Break Log	35
Fig 5.7	Add Attendance Page	36
Fig 5.8	Attendance List – Attendance Overview	37
Fig 5.9	Apply for Leave Page	38
Fig 5.10	Leave Records – My Leave Listing	39
Fig 5.11	Task Detail Page – Completed Task with Work Sessions	40
Fig 5.12	Admin Dashboard – Overview, Manual Work Log	41
Fig 5.13	Admin Dashboard – Manual Break Log and Work Logs Table	42
Fig 5.14	Admin Dashboard – Today's Manual Break Logs Table	43
Fig 5.15	Admin – Add Employee Form	44
Fig 5.16	Admin – Employee Directory Listing	45
Fig 5.17	Admin – Employee Attendance List	46
Fig 5.18	Admin – Attendance Details per Employee	47
Fig 5.19	Admin – Leave Management and Approval	48
Fig 5.20	Admin – All Projects Listing	49
Fig 5.21	Admin – Add New Project Form	50
Fig 5.22	Admin – Add New Task (Assignment Section)	51
Fig 5.23	Admin – Add New Task (Task Details and Priority)	52
Fig 5.24	Admin – Add New Task (Schedule and Duration)	53
Fig 5.25	Admin – Employee History Overview	54
Fig 5.26	Admin – Company Holidays Management	55
Fig 5.27	Admin – Add New Announcement Form	56
Fig 5.28	Admin – Announcements Listing	57

Figure No.	Figure Title	Page No.
Fig 5.29	Admin – Change Password Page	58

LIST OF TABLES

Table No.	Table Title	Page No.
Table 1.1	Organization Roles and Responsibilities	4
Table 2.1	Technical Tools and Technologies Used	6
Table 2.2	Week-wise Internship Activities	8
Table 3.1	Project Effort Estimation	17
Table 3.2	Project Gantt Chart	19
Table 4.1	Functional Requirements of the System	22
Table 4.2	Non-Functional Requirements	23
Table 4.3	Main Modules and Their Descriptions	28
Table 4.4	Hardware and Software Selection	29
Table 5.1	Database Table Overview	30
Table 6.1	Implementation Environment Specifications	43
Table 6.2	Comparison: Manual System vs. CRM EMS	48
Table 7.1	Test Cases – Employee Login Module	50
Table 7.2	Test Cases – Attendance Module	51
Table 7.3	Test Cases – Leave Management Module	52
Table 7.4	Test Cases – Task Management Module	53
Table A.1	Technology Stack Summary	62
Table A.2	Internship Details	63

LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

Abbreviation	Full Form
CRM	Customer Relationship Management
EMS	Employee Management System
UI	User Interface
UX	User Experience
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
PHP	Hypertext Preprocessor
MySQL	My Structured Query Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
SSL	Secure Sockets Layer
SDLC	Software Development Life Cycle
RBAC	Role-Based Access Control
SOU	Silver Oak University
ASOIT	Aditya Silver Oak Institute of Technology
REST	Representational State Transfer
UAT	User Acceptance Testing

TABLE OF CONTENTS

Acknowledgement	iv
Abstract	v
List of Figures	vi
List of Tables	vii
List of Abbreviations	viii
Table of Contents	ix
CHAPTER 1 OVERVIEW OF THE COMPANY	1
1.1 About Vision Technolabs	1
1.2 History and Background	2
1.3 Products and Scope of Work	3
1.4 Organization Structure	4
CHAPTER 2 OVERVIEW OF DEPARTMENT AND INTERNSHIP WORK	5
2.1 Department Overview	5
2.2 Technical Specifications of Tools Used	6
2.3 Workflow and Process Layout	7
2.4 Internship Activities Details	8
CHAPTER 3 INTRODUCTION TO PROJECT AND PROJECT MANAGEMENT	10
3.1 Project Summary	10

3.2 Purpose	11
3.3 Objectives	12
3.4 Scope of the Project	13
3.5 Technology and Literature Review	14
3.6 Project Planning	16
3.6.1 Development Approach and Justification	16
3.6.2 Effort, Time and Cost Estimation	17
3.6.3 Roles and Responsibilities	18
3.7 Project Scheduling – Gantt Chart	19
CHAPTER 4 SYSTEM ANALYSIS	20
4.1 Study of Current System	20
4.2 Problems and Weaknesses	21
4.3 Requirements of New System	22
4.4 System Feasibility	24
4.5 Activity / Process in Proposed System	26
4.6 Features of New System	27
4.7 Main Modules and Components	28
4.8 Hardware / Software Selection	29
CHAPTER 5 SYSTEM DESIGN	30
5.1 System Design and Methodology	30
5.2 Input / Output and Interface Design	30
5.2.1 Employee Side – Interface Screenshots	30
5.2.2 Admin Side – Interface Screenshots	41

5.2.3 Access Control and Security	59
CHAPTER 6 IMPLEMENTATION	61
6.1 Implementation Platform / Environment	61
6.2 Technology / Modules Specification	62
6.3 Findings / Results / Outcomes	65
6.4 Result Analysis and Deliberations	67
CHAPTER 7 TESTING	69
7.1 Testing Plan / Strategy	69
7.2 Test Results and Analysis	70
CHAPTER 8 CONCLUSION AND DISCUSSION	75
8.1 Overall Analysis of Project Viabilities	75
8.2 Photographs and Mentor Visit	76
8.3 Dates of Continuous Evaluation	76
8.4 Problems Encountered and Solutions	77
8.5 Summary of Internship Work	78
8.6 Limitations and Future Enhancements	79
REFERENCES / BIBLIOGRAPHY	81
APPENDIX	83

CHAPTER 1 OVERVIEW OF THE COMPANY

1.1 ABOUT VISION TECHNOLABS

Vision Technolabs is a dynamic and innovative software development company based in Ahmedabad, Gujarat, India. The company specializes in delivering cutting-edge digital solutions to businesses across various sectors, including enterprise software development, web application development, mobile application development, and IT consulting services. The organization operates its internal HR and employee management portal through the domain people.visiontechnolabs.com.

Vision Technolabs operates with a mission to empower businesses through technology, delivering tailored software solutions that drive growth, efficiency, and innovation. The company believes in fostering a collaborative work environment where fresh talent is nurtured alongside experienced professionals.

1.2 HISTORY AND BACKGROUND

Vision Technolabs was established with the vision of bridging the gap between technology and business needs of modern enterprises. Over the years, the company has grown from a small development studio into a full-service IT solutions provider. The company has built a portfolio of successful web and mobile applications catering to industries ranging from healthcare and education to e-commerce and enterprise resource planning.

The company places strong emphasis on research and development, continuously adopting modern frameworks and technologies to deliver state-of-the-art products to its clients. The internship program at Vision Technolabs is structured to give students practical exposure to real-world software development methodologies.

1.3 PRODUCTS AND SCOPE OF WORK

Vision Technolabs offers a comprehensive range of services and products:

- Custom Web Application Development using PHP, Laravel, React.js, and Node.js
- Mobile Application Development for Android and iOS platforms
- CRM and ERP System Development for enterprise clients
- UI/UX Design and Front-end Development
- Database Design and Administration using MySQL and PostgreSQL
- IT Consulting and Digital Transformation Services
- API Development and Third-party Integration
- Cloud Deployment and DevOps Services

During this internship, the primary work undertaken was the full-stack development of the CRM Employee Management System, which serves as an internal HR portal.

1.4 ORGANIZATION STRUCTURE

Vision Technolabs follows a lean and agile organizational structure headed by the Founder and CEO, supported by Senior Developers, Project Managers, UI/UX Designers, QA Engineers, and Business Development Executives. Interns work under the direct supervision of senior developers and project leads.

Table 1.1 Organization Roles and Responsibilities

Role	Responsibility
CEO / Founder	Overall company vision and strategic direction
Project Manager	Project planning, team coordination and delivery
Senior Developer	Architecture design, code review and mentorship
UI/UX Designer	Interface design and user experience prototyping
Junior Developer / Intern	Feature implementation, testing and documentation
QA Engineer	Quality assurance and test execution

CHAPTER 2 OVERVIEW OF DEPARTMENT AND INTERNSHIP WORK

2.1 DEPARTMENT OVERVIEW

The internship was carried out in the Software Development Department of Vision Technolabs. This department is responsible for the design, development, testing, and deployment of all web and mobile applications. The department follows an Agile methodology with weekly sprint planning, daily stand-up meetings, and iterative development cycles.

2.2 TECHNICAL SPECIFICATIONS OF TOOLS USED

Table 2.1 Technical Tools and Technologies Used

Tool / Technology	Version / Type	Purpose
PHP	8.x	Backend server-side scripting
MySQL	8.x	Relational database management
HTML5 / CSS3	Latest	Web page structure and styling
JavaScript / jQuery	ES6+	Client-side interactivity
Bootstrap	5.x	Responsive UI framework
XAMPP / Apache	Local Server	Local development environment
Visual Studio Code	Latest	Primary code editor
Git / GitHub	Latest	Version control and collaboration
Postman	v10.x	API testing and debugging

2.3 WORKFLOW AND PROCESS LAYOUT

The internship workflow followed a structured SDLC approach:

- **Requirements Gathering:** Detailed discussions with the project mentor to understand functional and non-functional requirements.
- **System Design:** Creating the database schema, wireframes, and module flow diagrams.
- **Front-end Development:** Designing and implementing all UI pages for Admin and Employee panels.
- **Back-end Development:** Implementing server-side logic using PHP and connecting with MySQL.
- **Integration and Testing:** Connecting front-end with back-end and performing unit and integration testing.
- **Review and Refinement:** Iterative feedback from the mentor leading to improvements and bug fixes.

2.4 INTERNSHIP ACTIVITIES DETAILS

Table 2.2 Week-wise Internship Activities

Period	Activities Performed
Week 1–2	Environment setup, understanding project requirements and technology stack. Completed task analysis.
Week 3–4	Database design and schema creation for all major modules.
Week 5–6	Front-end development – Employee Login, Dashboard with active session timer, Attendance module.
Week 7–8	Development of Leave Management module with multi-step form and listing.
Week 9–10	Task Management module – task timer, task completion tracking, work session log.
Week 11–12	Admin Panel development – employee management, attendance, leave approval, task assignment.
Week 13–14	Testing, debugging, UI refinements, and final documentation.

CHAPTER 3 INTRODUCTION TO PROJECT AND PROJECT MANAGEMENT

3.1 PROJECT SUMMARY

The CRM Employee Management System (EMS) is a full-stack web-based application developed to automate and manage the complete lifecycle of employee activities within an organization. The system provides a centralized platform for attendance management, task tracking, leave management, break time monitoring, and organizational communication. It is divided into two key panels – the **Admin Panel** and the **Employee Panel** – each providing role-specific functionalities with secure session-based authentication.

3.2 PURPOSE

The primary purpose is to eliminate manual, paper-based or Excel-based processes traditionally used for managing employee attendance, leave records, and task tracking. This system provides a single, unified digital platform where both employees and administrators can manage their daily activities efficiently, transparently, and in real time.

3.3 OBJECTIVES

- To provide a secure, role-based login system for Admins and Employees.
- To enable employees to track daily working hours through an active session timer.
- To allow employees to mark daily attendance as Present or Absent.

- To facilitate task assignment by administrators and task tracking by employees with dedicated timers.
- To implement a leave management system with approval workflow.
- To track employee break durations through an automated break logging mechanism.
- To provide administrators with comprehensive work-history reports.
- To enable administrators to manually add work and break logs for employees.
- To manage company holidays and announcements through the portal.
- To display team member birthdays and upcoming holidays on the employee dashboard.

3.4 SCOPE OF THE PROJECT

3.4.1 What the System Can Do

- Manage employee profiles with photo and personal details
- Real-time active session timer for tracking daily work hours
- Attendance marking and monthly attendance report generation
- Leave application and approval workflow
- Task assignment with deadlines, priority levels, and progress tracking
- Break time logging with start/end time and duration tracking
- Holiday management with upcoming holiday visibility
- Company-wide announcements to all employees
- Admin dashboard with employee summary statistics
- Password update for both Admin and Employee accounts

3.4.2 Limitations (What It Cannot Do)

- Does not support payroll processing or salary calculation.
- Does not support mobile push notifications in the current version.
- Does not include advanced analytics or ML-based insights.
- Does not support biometric device integration currently.

3.5 TECHNOLOGY AND LITERATURE REVIEW

PHP (Hypertext Preprocessor) is one of the most widely used server-side scripting languages for web development. Its seamless integration with MySQL makes it ideal for data-driven web applications.

MySQL is an open-source relational database management system known for its reliability, scalability, and ease of use, supporting complex queries and transactions suitable for enterprise applications.

Bootstrap 5 is a popular front-end framework providing responsive design components, enabling mobile-friendly user interfaces with minimal custom CSS. The CRM Employee Management System draws inspiration from leading HR platforms such as BambooHR, Zoho People, and Darwinbox, while adapting functionalities to Vision Technolabs.

3.6 PROJECT PLANNING

3.6.1 Development Approach and Justification

The **Agile Development Methodology** was adopted. Each sprint of two weeks focused on specific modules with review sessions after each sprint, allowing iterative improvements based on mentor feedback.

3.6.2 Effort, Time and Cost Estimation

Table 3.1 Project Effort Estimation

Module	Estimated Hours	Actual Hours
Requirements & Design	20 hrs	22 hrs
Employee Panel Development	60 hrs	65 hrs
Admin Panel Development	50 hrs	55 hrs
Database Design & Integration	20 hrs	18 hrs
Testing & Debugging	20 hrs	25 hrs
Documentation	15 hrs	15 hrs
Total	185 hrs	200 hrs

3.6.3 Roles and Responsibilities

- **Joshi yug Mukeshbhai (Intern Developer):** Full-stack development, database design, UI implementation, and report preparation.
- **Mr. Nayan Meghwal (Internal Guide):** Academic guidance and evaluation.
- **Vision Technolabs Mentor:** Project supervision, technical guidance, and feedback.

3.7 PROJECT SCHEDULING – GANTT CHART

Table 3.2 Project Gantt Chart

Task / Week	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Requirements & Setup	■	■												
Database Design			■	■										
Employee Panel – UI					■	■								
Leave & Task Module							■	■						
Admin Panel									■	■	■	■		
Testing & Debugging													■	■
Documentation													■	■

CHAPTER 4 SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEM

Prior to this system, Vision Technolabs relied on manual methods for managing employee processes. Attendance was tracked using physical registers or Excel sheets. Leave applications were submitted via email. Task assignments were communicated through messaging apps without any formal tracking. Work hour tracking was entirely absent, making it impossible to accurately measure individual employee productivity.

4.2 PROBLEMS AND WEAKNESSES OF CURRENT SYSTEM

- Manual attendance records are prone to human errors and manipulation.
- No mechanism to accurately track daily work hours and break times.
- Leave management through email lacks transparency and proper approval workflows.
- No centralized platform for task assignment and progress monitoring.
- Difficulty in generating attendance and work reports for performance reviews.
- Lack of real-time visibility into employee availability and status.
- Holiday and announcement information is scattered across different channels.
- Data is not stored in a structured database, making retrieval and analysis difficult.

4.3 REQUIREMENTS OF NEW SYSTEM

4.3.1 Functional Requirements

Table 4.1 Functional Requirements of the System

SR No.	Functional Requirement	Priority
FR-01	Secure role-based login for Admin and Employee	High
FR-02	Employee can start/stop/reset work timer	High
FR-03	Employee can mark attendance as Present or Absent	High
FR-04	Employee can log break start and end times	High
FR-05	Admin can assign tasks with deadlines and priority	High
FR-06	Employee can start/stop task timer and mark as Done	High
FR-07	Employee can apply for full-day or half-day leave	High
FR-08	Admin can approve or reject leave applications	High
FR-09	Admin can manually add work log and break log entries	Medium
FR-10	Admin can add/edit/delete employee profiles	High
FR-11	Admin can manage company holiday calendar	Medium
FR-12	Admin can post and manage company announcements	Medium
FR-13	Admin can create projects and list all projects	High
FR-14	Both Admin and Employee can update their passwords	Medium

4.3.2 Non-Functional Requirements

SR No.	Non-Functional Requirement
NFR-01	Pages must load within 3 seconds under normal network conditions.
NFR-02	Compatible with all modern browsers (Chrome, Firefox, Edge).
NFR-03	Responsive design for desktop and tablet screens.
NFR-04	All passwords stored in encrypted (BCRYPT hashed) format.
NFR-05	Session-based authentication prevents unauthorized access.
NFR-06	Regular database backups to prevent data loss.
NFR-07	Intuitive UI requiring minimal training for new users.

4.4 SYSTEM FEASIBILITY

4.4.1 Technical Feasibility

The system is built entirely using open-source technologies (PHP, MySQL, Bootstrap, JavaScript), significantly reducing development cost. Deployment is possible on affordable shared hosting or a basic VPS server.

4.4.2 Operational Feasibility

The system directly addresses the operational need for workforce efficiency. By automating attendance tracking, task management, and leave processing, it saves significant time for employees and administrators.

4.4.3 Economic Feasibility

The entire system was developed within the 14-week internship period with zero software licensing costs, demonstrating strong economic feasibility.

4.5 ACTIVITY / PROCESS IN PROPOSED SYSTEM

Upon logging in, employees are directed to their personalized dashboard where they can start their work timer, view their task board, check leave status, and access upcoming

holidays and announcements. The administrator gets a comprehensive overview of all employee activities from a single interface and can manage every aspect of the workforce.

4.6 FEATURES OF NEW / PROPOSED SYSTEM

- Dual-panel architecture with role-based access control
- Real-time active work session timer with start, break, and reset controls
- Digital attendance marking with daily status tracking
- Multi-step leave application form with instant admin notifications
- Task management board with priority levels, deadlines, and time tracking
- Comprehensive break log with reason, start time, end time, and duration
- Admin work-history reports for individual employee performance analysis
- Project management – add projects, assign tasks to employees
- Holiday calendar management with employee-visible upcoming holidays
- Company-wide announcement system with listing and search
- Manual work log and break log entry by admin for missed timers
- Password management for both Admin and Employee accounts

4.7 MAIN MODULES AND COMPONENTS

Table 4.3 Main Modules and Their Descriptions

Module Name	Description
Authentication Module	Secure login/logout for Admin and Employee roles.
Employee Dashboard	Work timer, task board, break activity, team members, birthdays.
Attendance Module	Daily attendance marking and monthly attendance history.
Leave Management Module	Leave application, listing, and admin approval/rejection workflow.
Task Management Module	Task assignment, task timer, completion, and work session logging.
Break Log Module	Records break sessions with duration tracking.
Project Management Module	Admin creates and manages projects, assigns tasks to employees.
Admin Dashboard Module	Overview of total employees, manual work/break log entry.
Employee Management	Admin adds, edits, and manages employee profiles.
Work History Module	Admin views complete work and break time history per employee.
Holiday Management	Admin manages company holidays visible to all employees.
Announcement Module	Admin posts and manages announcements for all employees.
Password Management	Secure password update for both Admin and Employee.

4.8 SELECTION OF HARDWARE / SOFTWARE AND JUSTIFICATION

Table 4.4 Hardware and Software Selection

Component	Selection	Justification
Backend Language	PHP 8.x	Widely supported, extensive docs, easy MySQL integration
Database	MySQL 8.x	Reliable, open-source, supports complex queries.
Frontend Framework	Bootstrap 5	Responsive components reduce custom CSS development
Scripting	JS / jQuery	Industry-standard client-side scripting.
Web Server	Apache (XAMPP)	Easy local development with PHP and MySQL bundled.
Code Editor	VS Code	Free, feature-rich, excellent PHP and web extensions.
Version Control	Git / GitHub	Industry-standard collaborative version control.

CHAPTER 5 SYSTEM DESIGN

5.1 SYSTEM DESIGN AND METHODOLOGY

The CRM Employee Management System is designed using a three-tier architecture: the Presentation Layer (front-end UI), the Application Layer (server-side PHP logic), and the Data Layer (MySQL database). This separation of concerns makes the application modular, maintainable, and scalable.

The database uses a normalized relational model to minimize redundancy and ensure referential integrity. Key tables include: users, attendance, leave_applications, tasks, work_sessions, break_logs, holidays, announcements, and employee_profiles.

Table 5.1 Database Table Overview

Table Name	Primary Key	Description
users	user_id	All user accounts (Admin and Employee) with roles
attendance	att_id	Daily attendance entries with date, time and status
leave_applications	leave_id	Leave requests with type, reason and approval status
tasks	task_id	Tasks assigned by admin with priority and deadline
work_sessions	session_id	Work timer sessions with start time, end time, duration
break_logs	break_id	Break sessions with reason, start, end and duration
holidays	holiday_id	Company holiday dates with names and descriptions
announcements	ann_id	Company announcements with title, content and date
employee_profiles	profile_id	Extended profile info including photo and designation

5.2 INPUT / OUTPUT AND INTERFACE DESIGN

5.2.1 Employee Side – Interface Screenshots

The following figures illustrate the key screens of the Employee side of the CRM Employee Management System as observed and used during the internship.

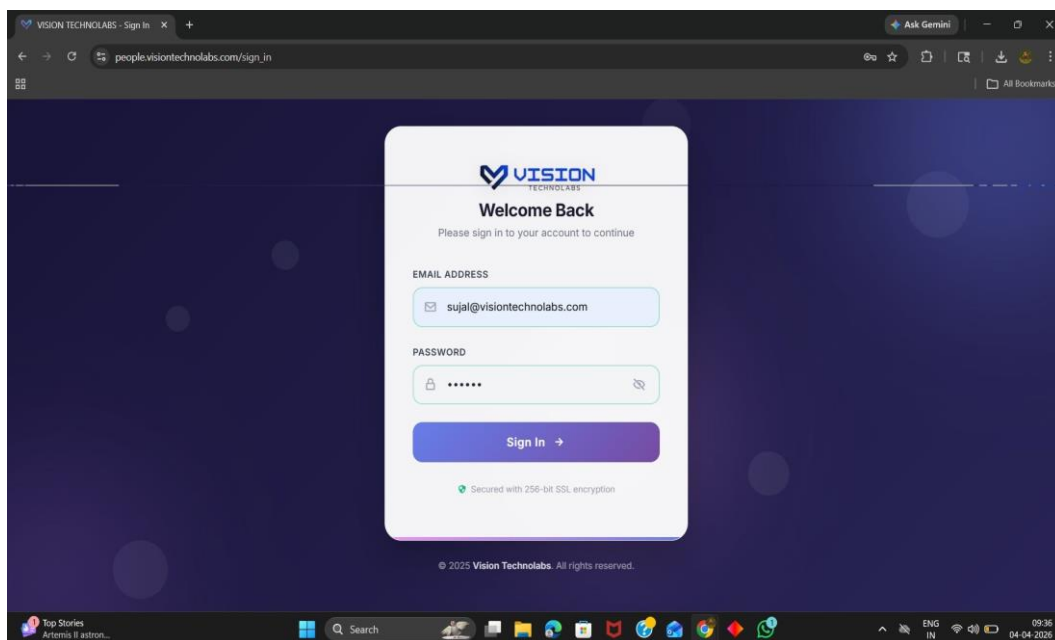


Fig 5.1 Employee Login Page

The login page provides a clean and secure authentication interface for the Vision Technolabs employee portal. It requires the employee's registered email address and password. The form is secured with 256-bit SSL encryption and hosted at people.visiontechnolabs.com/sign_in.

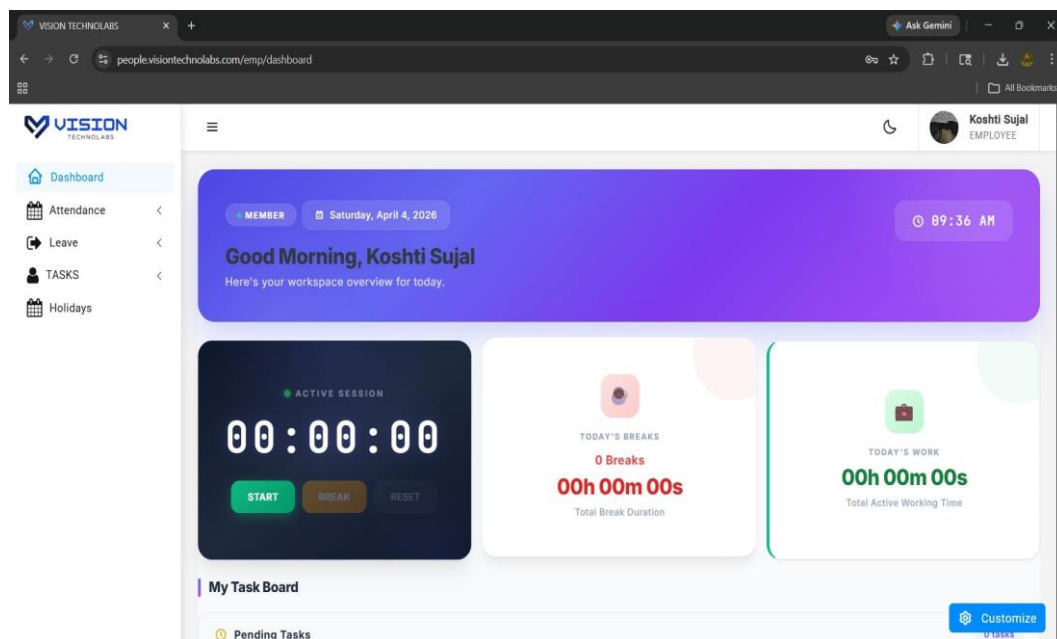


Fig 5.2 Employee Dashboard – Active Session Timer

The Employee Dashboard greets the logged-in employee with a personalized welcome banner showing the current date and time. The Active Session Timer card prominently displays the running work timer with START, BREAK, and RESET buttons. The 'Today's Breaks' card shows break count and total break duration; 'Today's Work' card shows total active working time.

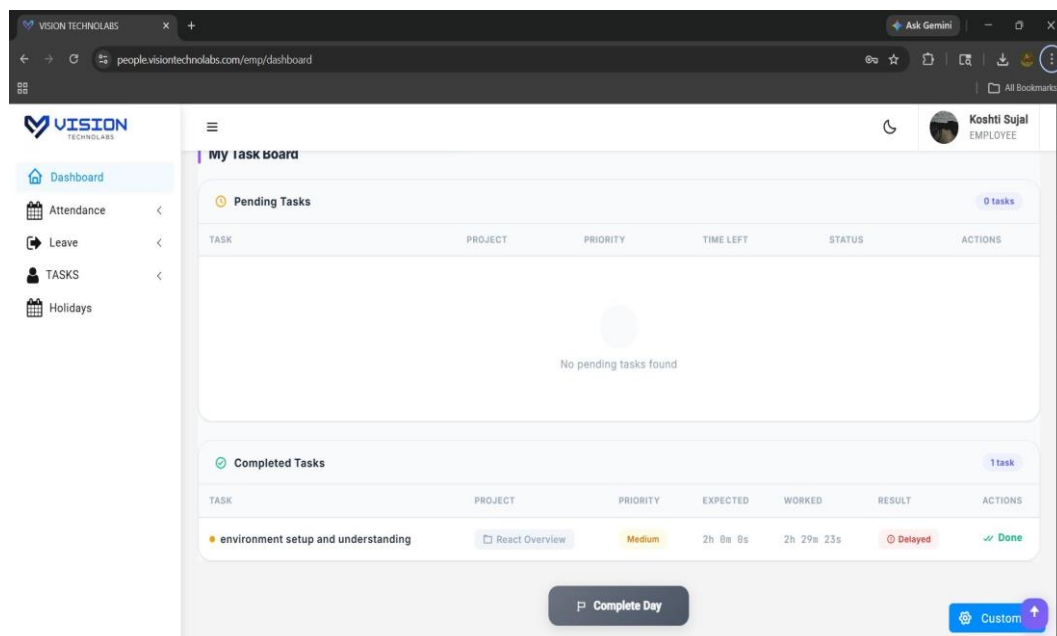


Fig 5.3 Employee Dashboard – Task Board (Pending Tasks)

The Task Board section displays all pending tasks assigned by the administrator. Each task entry shows the task name, associated project, priority level (High/Medium/Low), time remaining, and current status. The 'No pending tasks found' state is shown when all assigned tasks are completed.

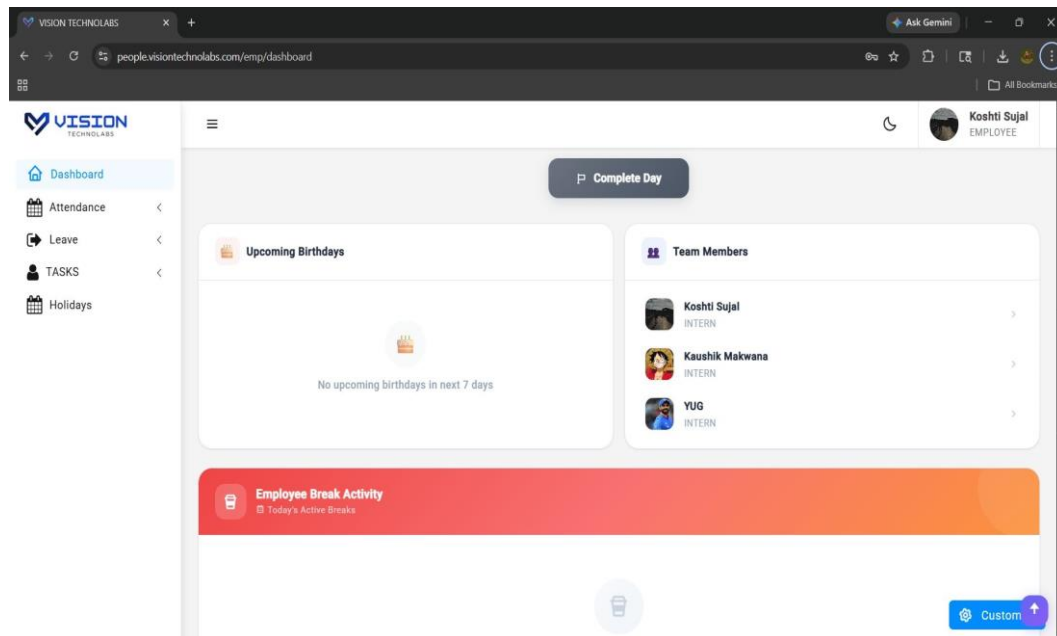


Fig 5.4 Employee Dashboard – Completed Tasks

The Completed Tasks section of the Task Board shows each completed task with its name, project, priority, expected time, actual worked time, result (On-Time/Delayed), and final status (Done). The 'Complete Day' button at the bottom formally closes the workday.

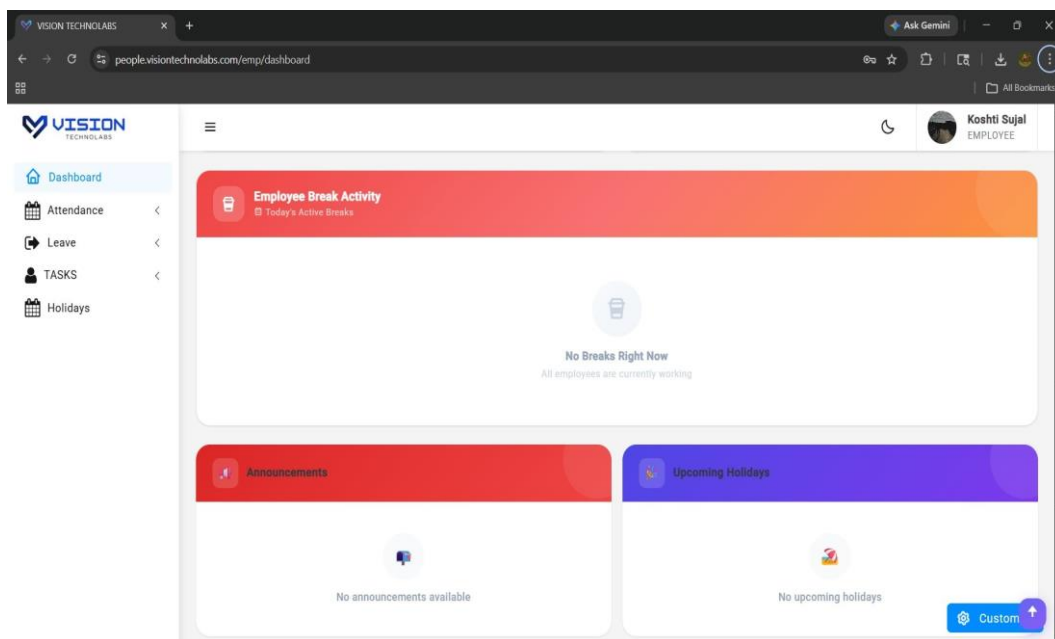


Fig 5.5 Employee Dashboard – Employee Break Activity

The 'Employee Break Activity' section shows all employees currently on break. When all employees are actively working, it displays 'No Breaks Right Now – All employees are currently working', providing real-time visibility into team availability.

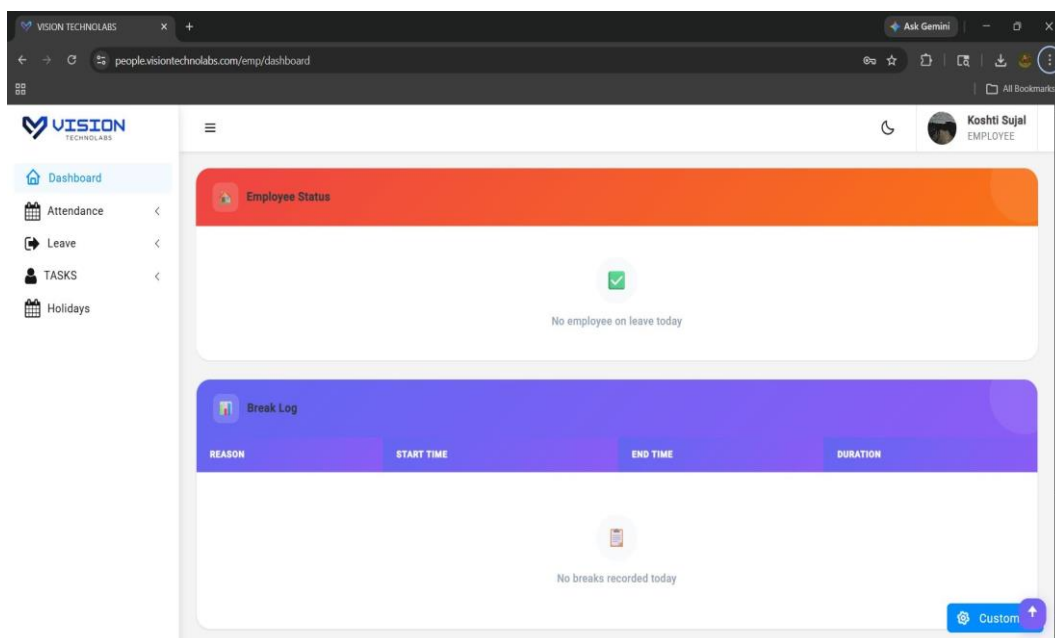


Fig 5.6 Employee Dashboard – Employee Status and Break Log

The 'Employee Status' card shows whether any team member is on leave. The 'Break Log' table records each break session with Reason, Start Time, End Time, and Duration. Announcements and Upcoming Holidays sections are also visible on this portion of the dashboard.

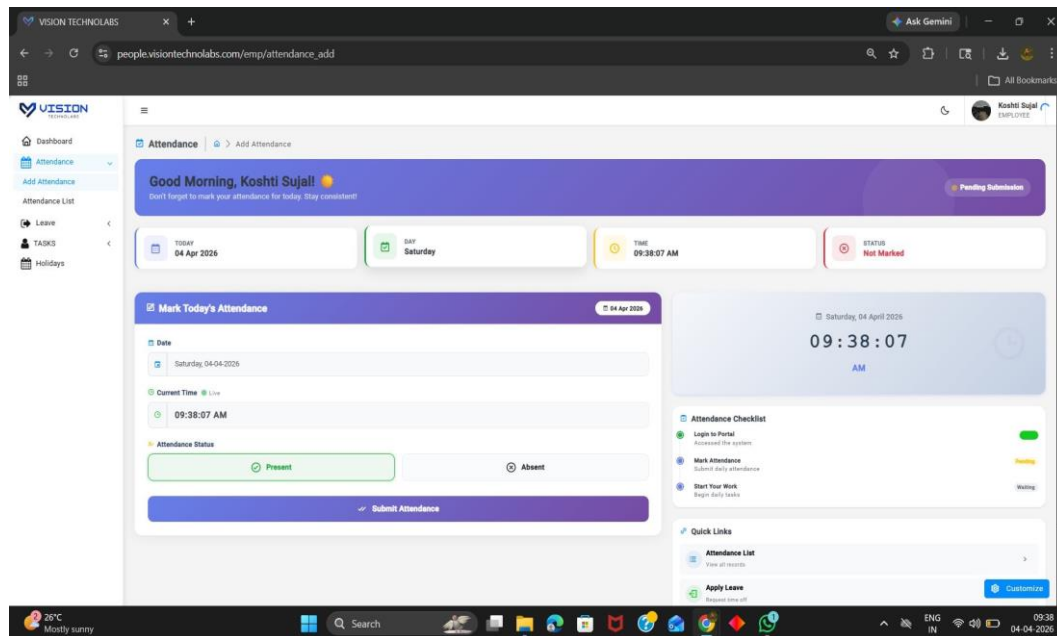


Fig 5.7 Add Attendance Page

The Add Attendance page allows employees to mark daily attendance as Present or Absent. It displays the current date, day, and live time. An Attendance Checklist guides employees through three daily steps: Login to Portal (completed), Mark Attendance (pending), and Start Your Work (waiting).

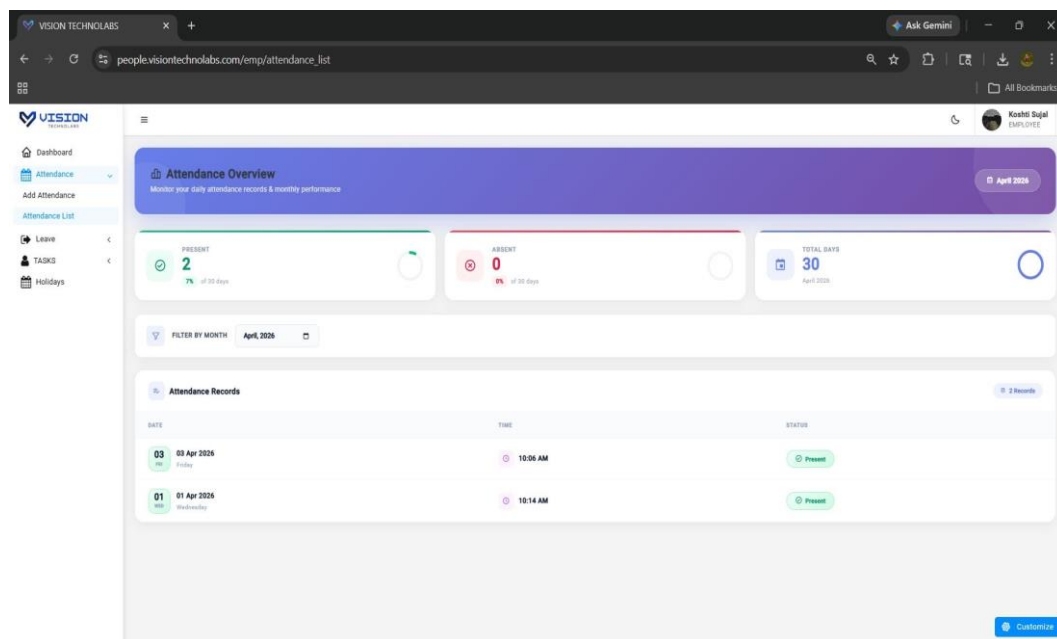


Fig 5.8 Attendance List – Attendance Overview

The Attendance List page provides a monthly Attendance Overview with summary cards showing Present days, Absent days, and Total Days in the month. The Attendance Records table lists each day's attendance entry with date, time of marking, and status (Present/Absent).

The screenshot displays the 'Apply for Leave' page. The sidebar is identical to the previous page. The main content area has a header 'Apply for Leave' with a sub-header 'Submit your leave request - Your manager will be notified instantly'. A date selector shows 'Submitting, 04 Apr 2024'. Below this, a 'Leave Application Form' section contains a progress bar with four steps: 1. EMPLOYEE, 2. REASON, 3. TYPE, and 4. REVIEW. A yellow warning box states: 'Leave requests must be submitted at least 5 days in advance. Emergency leave requests require manager approval.' The form includes a dropdown for 'EMPLOYEE' (Kushi Sagar), a date field for 'LEAVE DATE' (04 May 2024), and a 'LEAVE TYPE' section with three options: 'Full Day' (marked as selected), '1st Half' (Morning Session), and '2nd Half' (Afternoon Session). A 'REASON' section with a text area is also present. A 'Submit Leave Request' button is at the bottom. On the right, a 'Leave Guidelines' section lists rules for Full Day Leave, 1st Half (Morning), and 2nd Half (Afternoon), along with an 'Approval Process' diagram.

Fig 5.9 Apply for Leave Page

The Apply for Leave page features a multi-step leave application form with four steps: Employee, Details, Type, and Submit. The employee selects the leave date, chooses the leave type (Full Day, 1st Half, or 2nd Half), and optionally provides a reason. Leave requests must be submitted at least 1 day in advance.

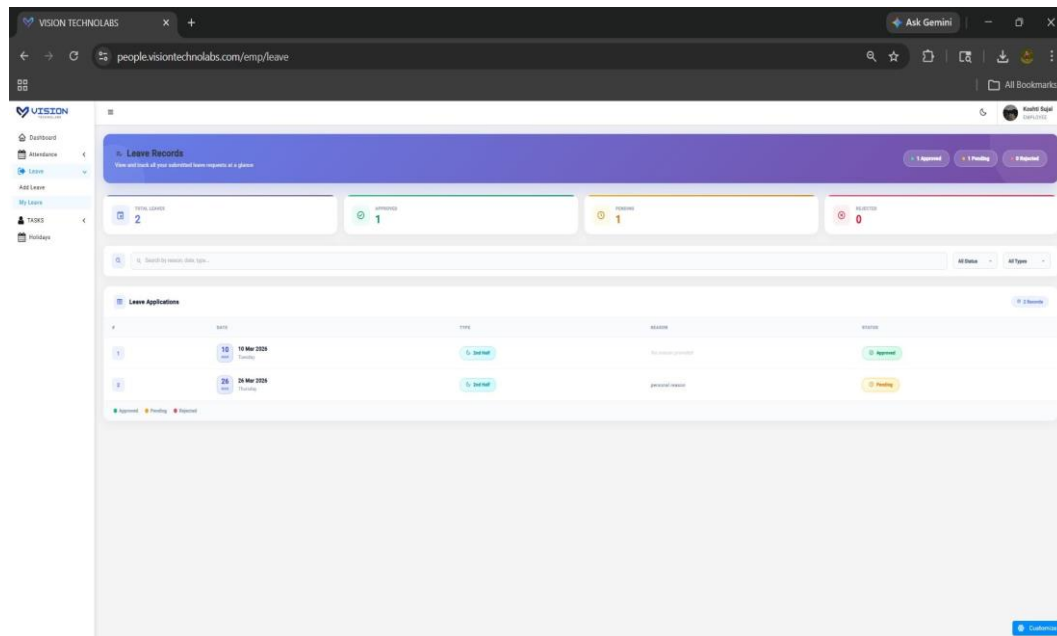


Fig 5.10 Leave Records – My Leave Listing

The Leave Records page shows all submitted leave applications with summary cards for Total Leaves, Approved, Pending, and Rejected counts. The table lists each application with date, type, reason, and approval status. Filter options by Status and Type allow easy navigation through leave history.

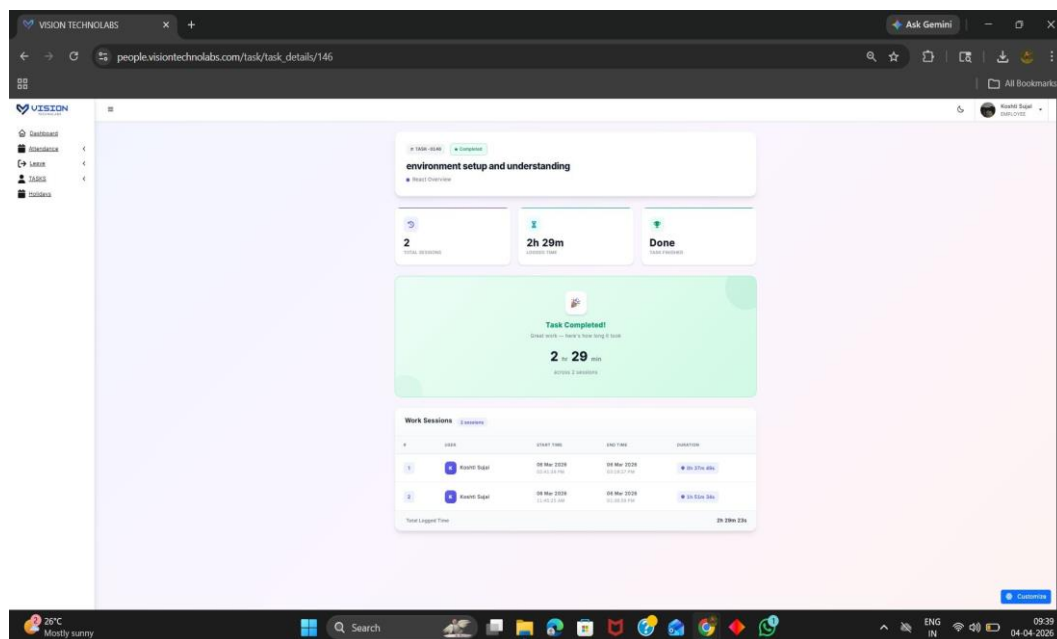


Fig 5.11 Task Detail Page – Completed Task with Work Sessions

The Task Detail page (TASK-0146: Environment Setup and Understanding) shows 2 Total Sessions, 2h 29m Logged Time, and Done status. The Work Sessions table details each session with user, start time, end time, and duration. Total Logged Time of 2h 29m 23s is shown at the bottom, confirming task completion (slightly delayed).

5.2.2 Admin Side – Interface Screenshots

The Admin Panel of the CRM Employee Management System provides comprehensive control over all employee management activities. The following figures illustrate all key screens of the Admin side as developed and deployed during the internship.

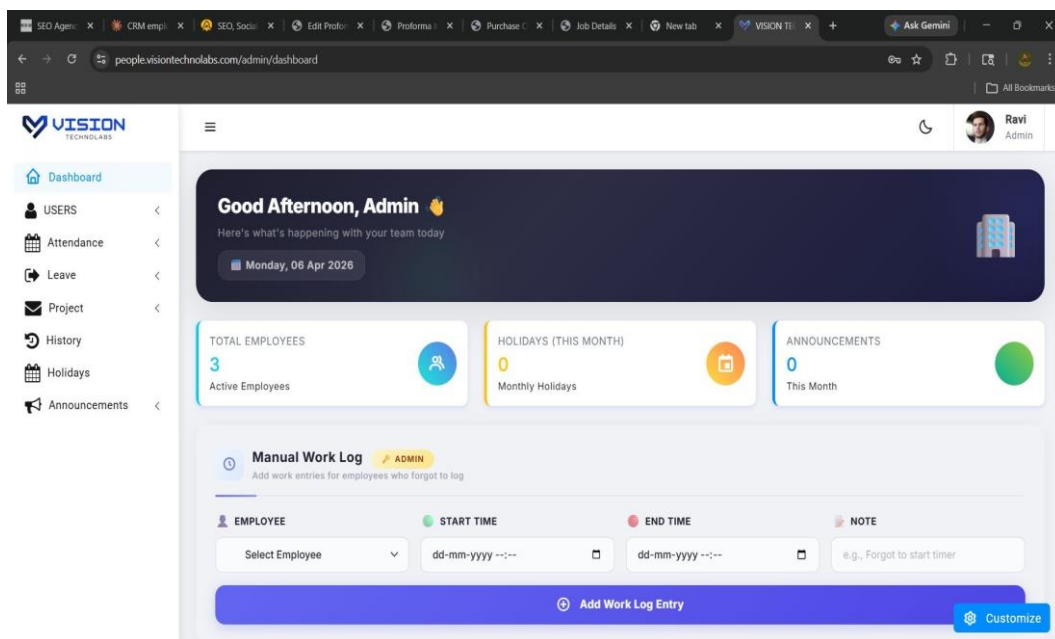


Fig 5.12 Admin Dashboard – Overview with Manual Work Log Entry

The Admin Dashboard (Fig 5.12) greets the administrator with a welcome banner showing today's date. Three summary cards display Total Employees (3 Active), Holidays This Month (0), and Total Announcements (0). Below these cards, the 'Manual Work Log' section — exclusively accessible to Admin — provides a form to add work log entries for employees who forgot to start their timer. The admin selects the employee, sets start and end time, and adds an optional note before clicking 'Add Work Log Entry'.

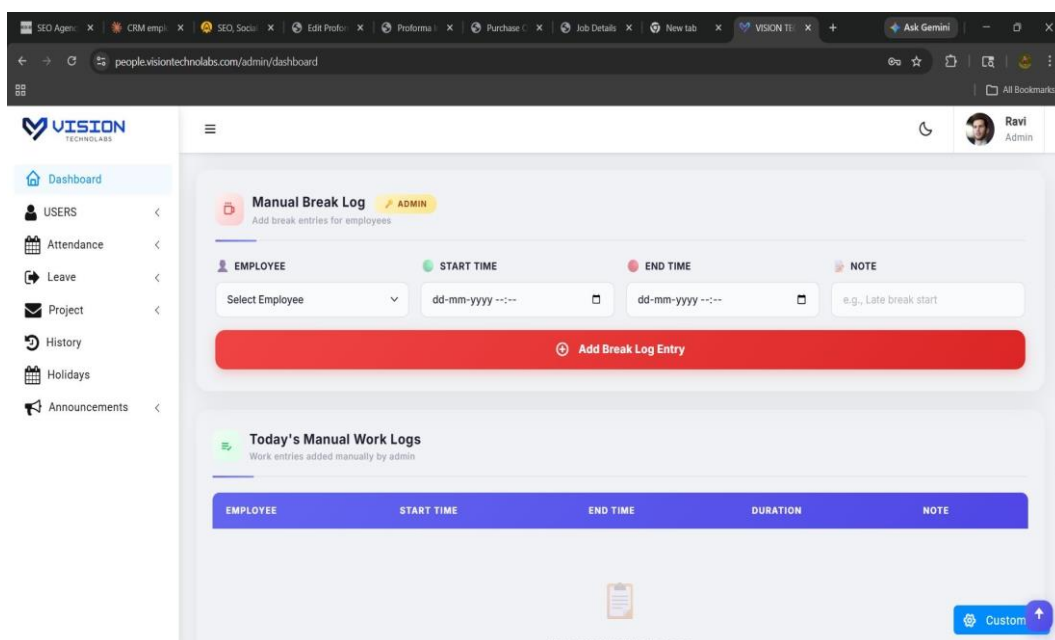


Fig 5.13 Admin Dashboard – Manual Break Log and Work Logs Table

Scrolling down the Admin Dashboard (Fig 5.13) reveals the 'Manual Break Log' section, where admin can add break entries for employees who missed logging their break. The form includes Employee selection, Start Time, End Time, and a Note field. Below, 'Today's Manual Work Logs' table lists all manually added work entries with columns for Employee, Start Time, End Time, Duration, and Note.

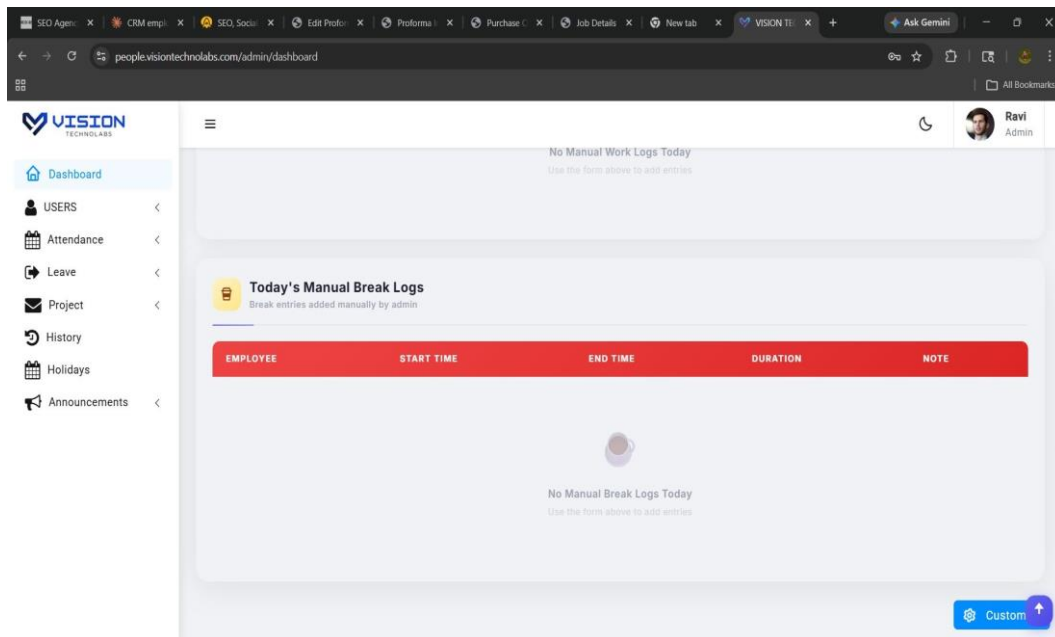


Fig 5.14 Admin Dashboard – Today's Manual Break Logs Table

The final section of the Admin Dashboard (Fig 5.14) shows 'Today's Manual Break Logs' — a table tracking all manually added break entries with Reason, Start Time, End Time, and Duration columns. When no manual logs have been added today, the state shows 'No Manual Break Logs Today – Use the form above to add entries'.

The screenshot shows the 'Add Employee' form in the VISION Technolabs admin interface. The form is divided into three sections: Personal Information, Account Details, and Location Details. The Personal Information section includes fields for Full Name, Phone Number, and a profile photo upload. The Account Details section includes fields for Username, Email Address, Password, and Confirm Password. The Location Details section includes a Country dropdown and an Address field. At the bottom, there are buttons for 'Create Employee' and 'Reset Form'.

Fig 5.15 Admin – Add Employee Form (Personal, Account and Location Details)

The Add Employee page (Fig 5.15) features a three-step form for creating new employee accounts: Step 1 (Personal) collects full name, phone number, and profile photo upload; Step 2 (Account) collects username, email address, password, and confirm password; Step 3 (Location) collects country and full address. The admin clicks 'Create Employee' to save the new employee account, which is then automatically added to the system database.

The screenshot shows the 'Employee Directory' page in the VISION Technolabs admin interface. The page displays a summary of employee statistics and a table of all employees. The statistics are as follows:

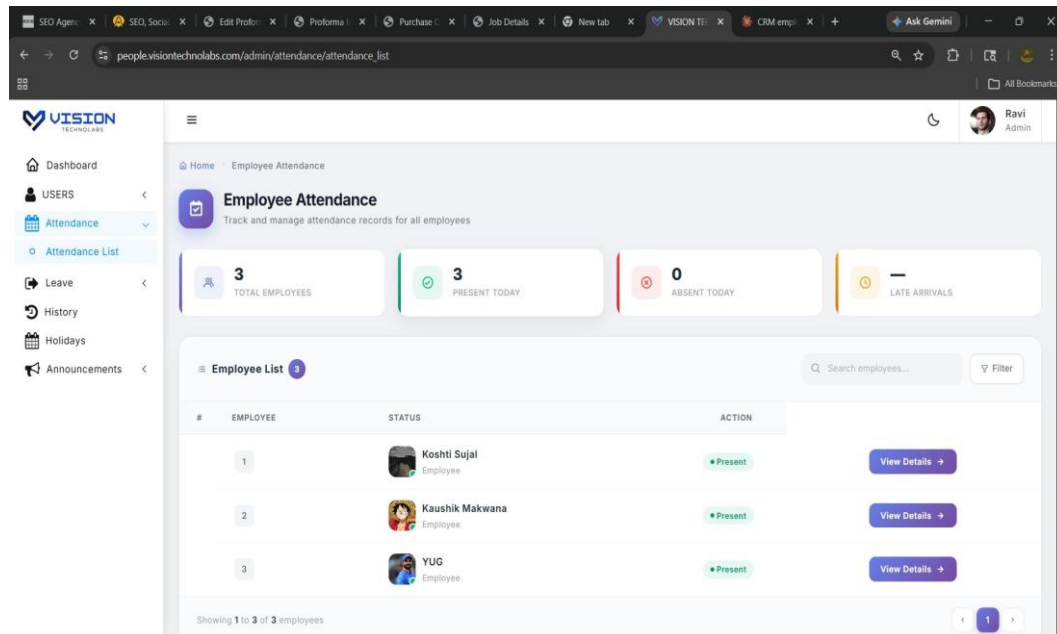
- 3 TOTAL EMPLOYEES
- 3 ACTIVE (100%)
- 0 NEW THIS MONTH
- 0 INACTIVE

The 'All Employees' table shows 3 records found:

#	EMPLOYEE	PHONE	EMAIL	STATUS	ACTIONS
1	KS Koshti Sujal ID: EMP-0016	7990911877	sujal@visiontechnolabs.com	ACTIVE	View, Edit, Delete
2	KM Kaushik Makwana ID: EMP-0017	9099780463	kaushik@visiontechnolabs.com	ACTIVE	View, Edit, Delete
3	Y YUG ID: EMP-0018	+919904300073	yug@visiontechnolabs.com	ACTIVE	View, Edit, Delete

Fig 5.16 Admin – Employee Directory Listing

The Employee Directory page (Fig 5.16) displays a comprehensive listing of all employees. Summary cards show Total Employees (3), Active (3 – 100%), New This Month (0), and Inactive (0). The All Employees table lists each employee with their photo avatar, name, Employee ID, phone number, email address, and Active status. Action buttons for each row allow the admin to View, Edit, or Delete employee records.

*Fig 5.17 Admin – Employee Attendance List*

The Employee Attendance page (Fig 5.17) gives the admin a real-time view of today's attendance for all employees. Summary cards show Total Employees (3), Present Today (3), Absent Today (0), and Late Arrivals (-). The Employee List table shows each employee's photo, name, designation, today's attendance status (Present/Absent), and a 'View Details' button. Search and Filter options are available for large teams.

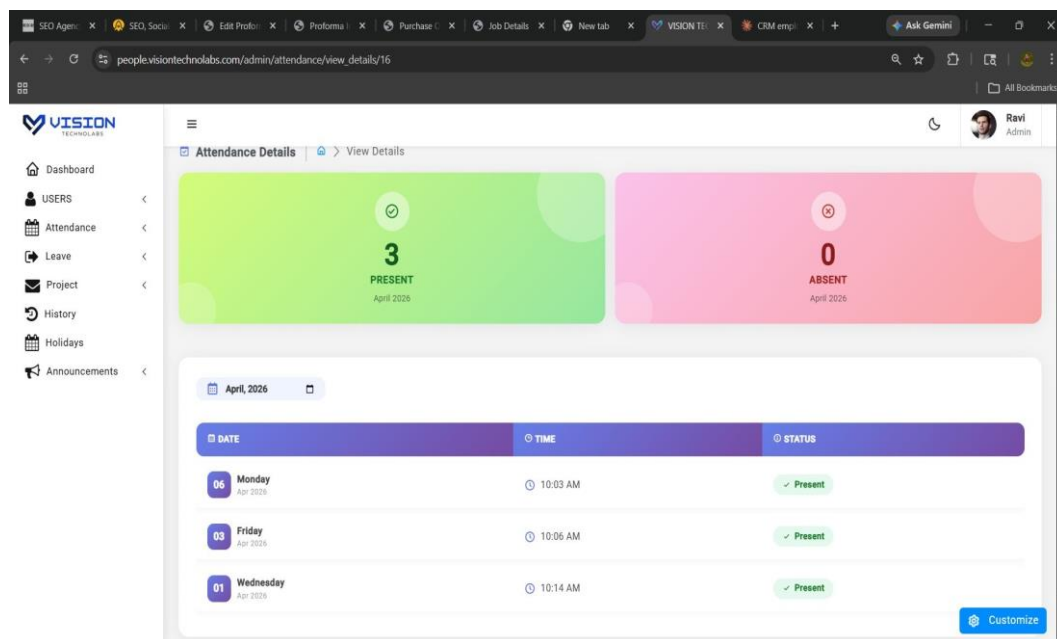


Fig 5.18 Admin – Attendance Details per Employee

Clicking 'View Details' on the Attendance List opens the Attendance Details page (Fig 5.18) for a specific employee. Two large summary cards show total Present days (3) and Absent days (0) for April 2026. A detailed Attendance Records table below lists each day's entry with DATE, TIME, and STATUS columns, filtered by the selected month.

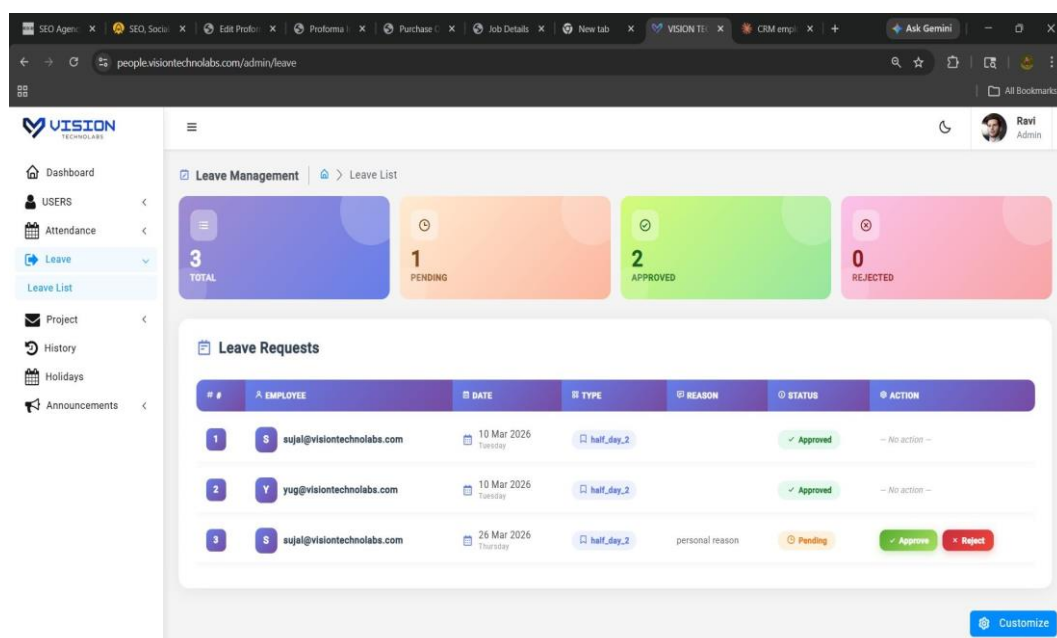


Fig 5.19 Admin – Leave Management and Approval

The Leave Management page (Fig 5.19) provides the admin with a complete view of all employee leave requests. Summary cards display Total (3), Pending (1), Approved (2), and Rejected (0) leave counts. The Leave Requests table lists each application with Employee email, Date, Type (half_day_2), Reason, Status, and Action buttons. The admin can approve or reject pending leaves directly from this page.

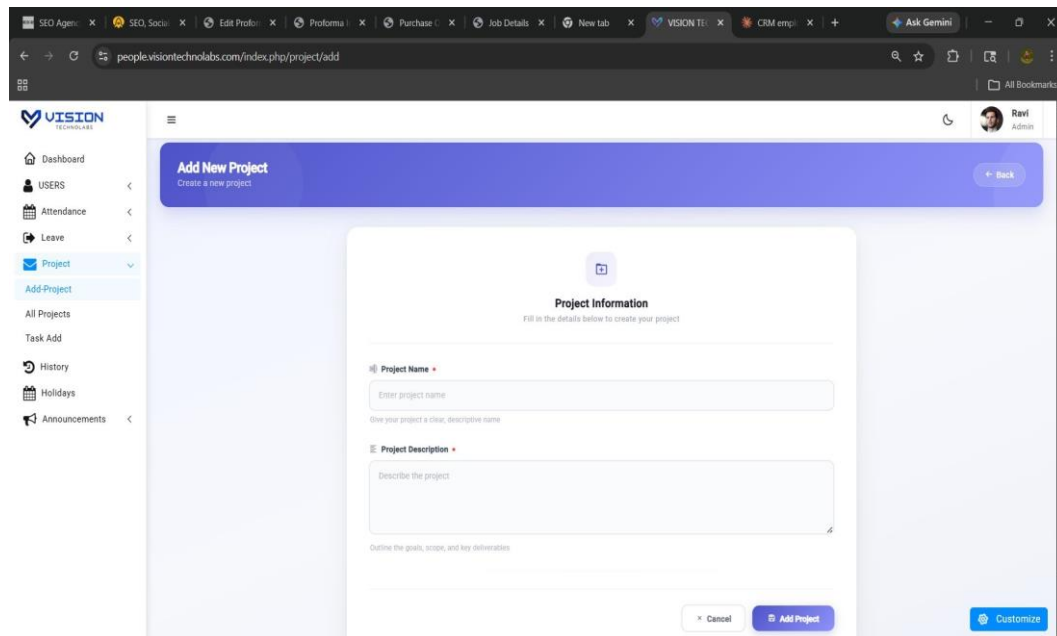


Fig 5.20 Admin – All Projects Listing

The My Projects page (Fig 5.20) lists all projects created in the system. Summary cards show Total Projects (3), Active Projects (0), Completed (0), and Total Tasks (0). The projects listed are SiteDesk, React Overview, and Lok Sangeet — all with Active status. Each project row has action buttons for viewing tasks, editing, and deleting the project.

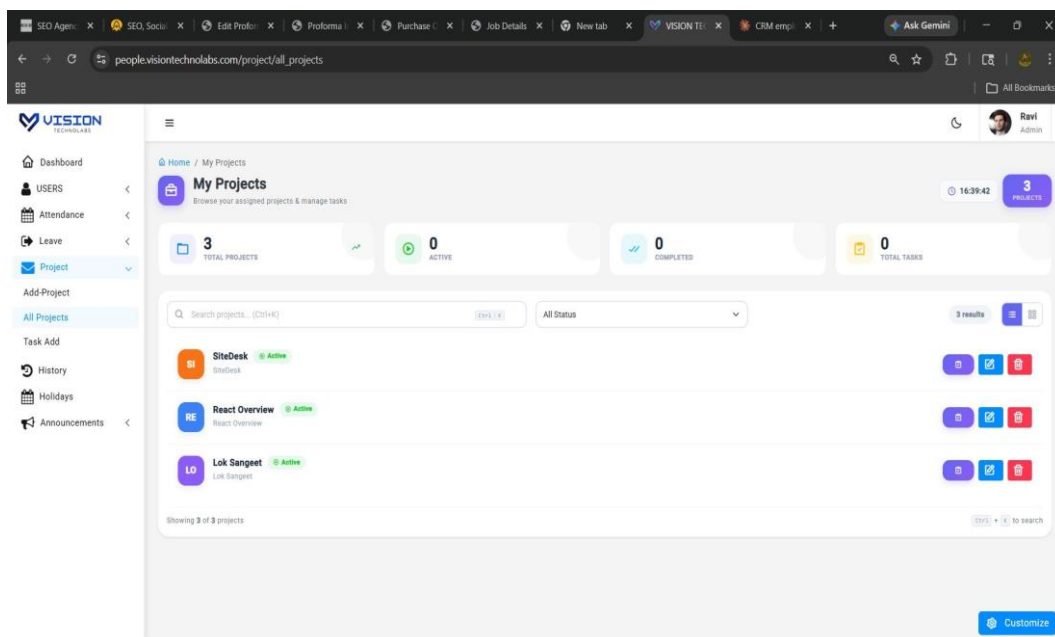


Fig 5.21 Admin – Add New Project Form

The Add New Project page (Fig 5.21) provides a simple form for creating new projects. The admin enters the Project Name and Project Description, then clicks 'Add Project'. Created projects appear in the All Projects listing and are available for task assignment.

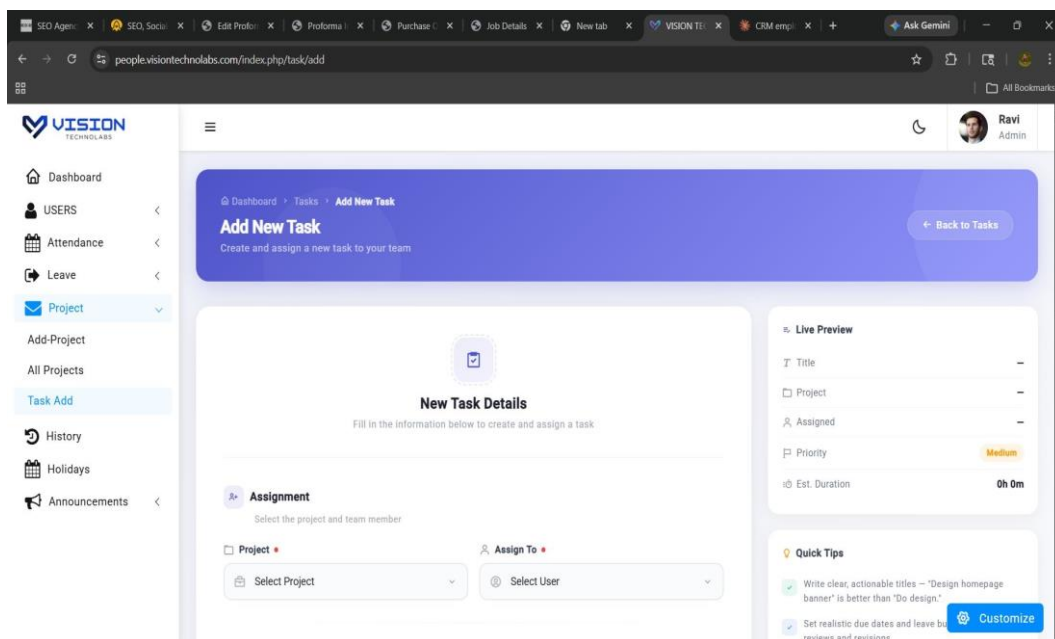


Fig 5.22 Admin – Add New Task (Assignment Section)

The Add New Task page (Fig 5.22) is a comprehensive form for creating and assigning tasks. The Assignment section lets admin select the Project and the team member (Assign To) from dropdown menus. A Live Preview panel on the right side shows real-time updates of the task details as the admin fills in the form, along with Quick Tips for effective task creation.

Fig 5.23 Admin – Add New Task (Task Details and Priority Level)

The Task Details section of the Add New Task form (Fig 5.23) requires the admin to enter a clear Task Title and an optional Description. The Priority Level section provides three selectable options: Low, Medium (default, shown highlighted in yellow), and High. The Live Preview panel continues to update as fields are completed.

Fig 5.24 Admin – Add New Task (Schedule and Duration)

The Schedule and Duration section (Fig 5.24) allows the admin to define the task timeline by setting the mandatory Start Date & Time and an optional Due Date. The Estimated Duration field accepts hours and minutes. The admin then clicks 'Add Task' to save. Keyboard shortcuts (Ctrl+Enter to submit, Escape to cancel) are also available.

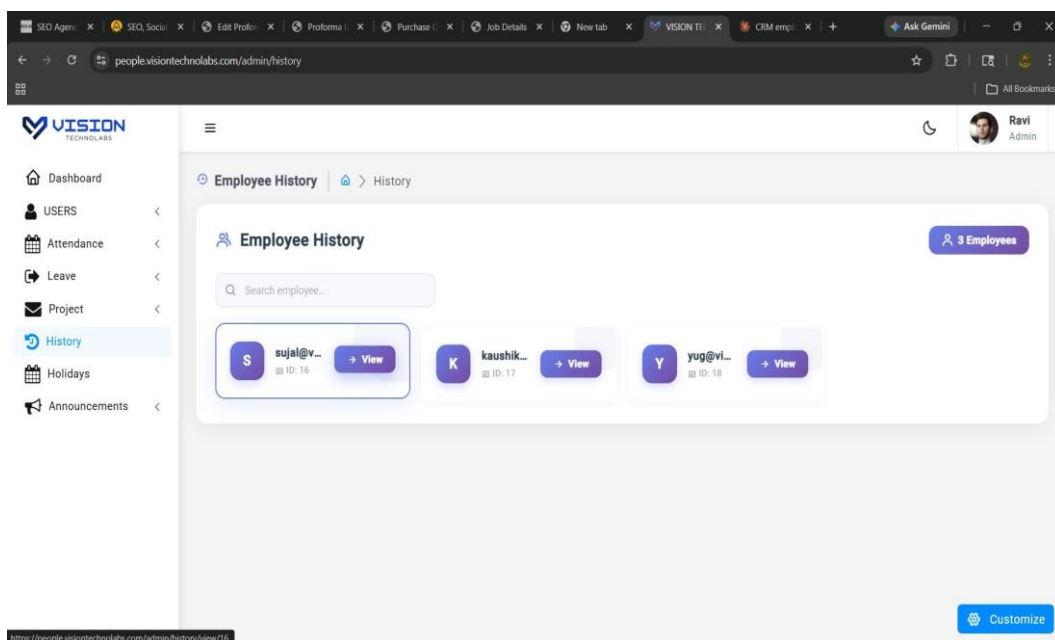


Fig 5.25 Admin – Employee History Overview

The Employee History page (Fig 5.25) provides a card-based view of all 3 employees.

Each card shows the employee's email (abbreviated), Employee ID (16, 17, 18), and a 'View' button. Clicking View opens the detailed work history for that employee, showing their total work time and break time across all sessions for admin review.

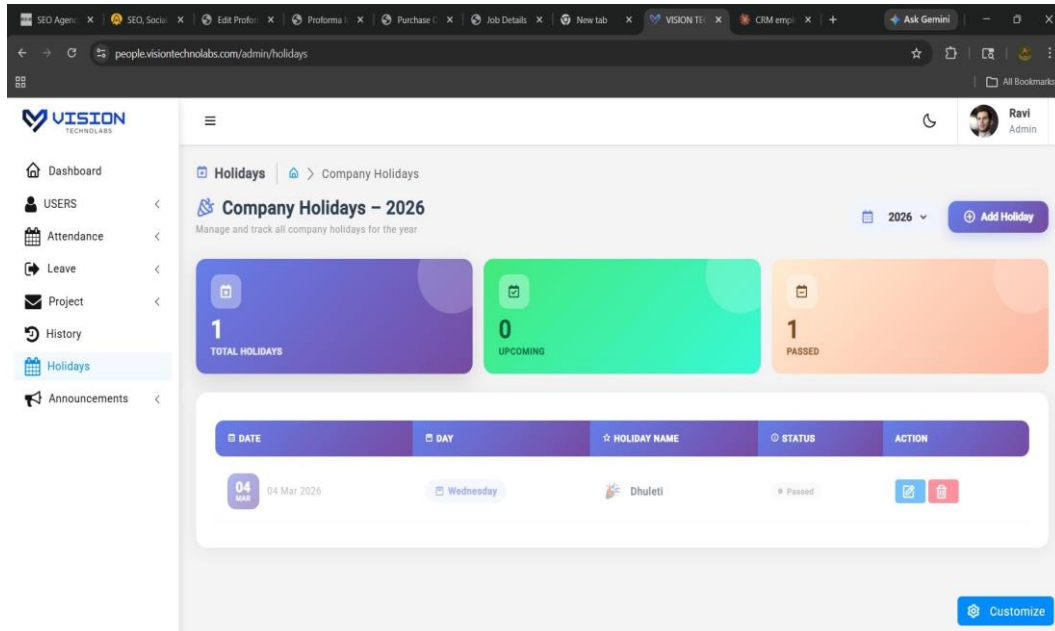


Fig 5.26 Admin – Company Holidays Management

The Company Holidays page for 2026 (Fig 5.26) shows all company holidays with summary cards for Total Holidays (1), Upcoming (0), and Passed (1). The holiday 'Dhuleti' on 04 Mar 2026 (Wednesday) appears with a Passed status. The admin can add new holidays using the 'Add Holiday' button and edit or delete existing holidays.

The screenshot shows the 'Create New Announcement' form in the VISION Technolabs admin dashboard. The form is titled 'Create New Announcement' and includes a sub-header 'Fill in the details below to publish a new announcement'. It contains two main input fields: 'Announcement Title' with a placeholder 'Enter a clear, concise title...' and a character count of 0/150; and 'Description' with a placeholder 'Write your announcement details here. Include all relevant information...' and a character count of 0/2000. Below the description field is a 'Preview' button. At the bottom of the form are 'Publish Announcement' and 'Cancel' buttons. The left sidebar shows the 'Announcements' menu item selected, and the top right shows the user 'Ravi Admin'.

Fig 5.27 Admin – Create New Announcement Form

The Create New Announcement page (Fig 5.27) provides a simple form for publishing company-wide announcements. The admin enters the Announcement Title (up to 150 characters) and a Description (up to 2000 characters). A Preview option allows the admin to review the announcement before publishing. Clicking 'Publish Announcement' makes it visible to all employees on their dashboard.

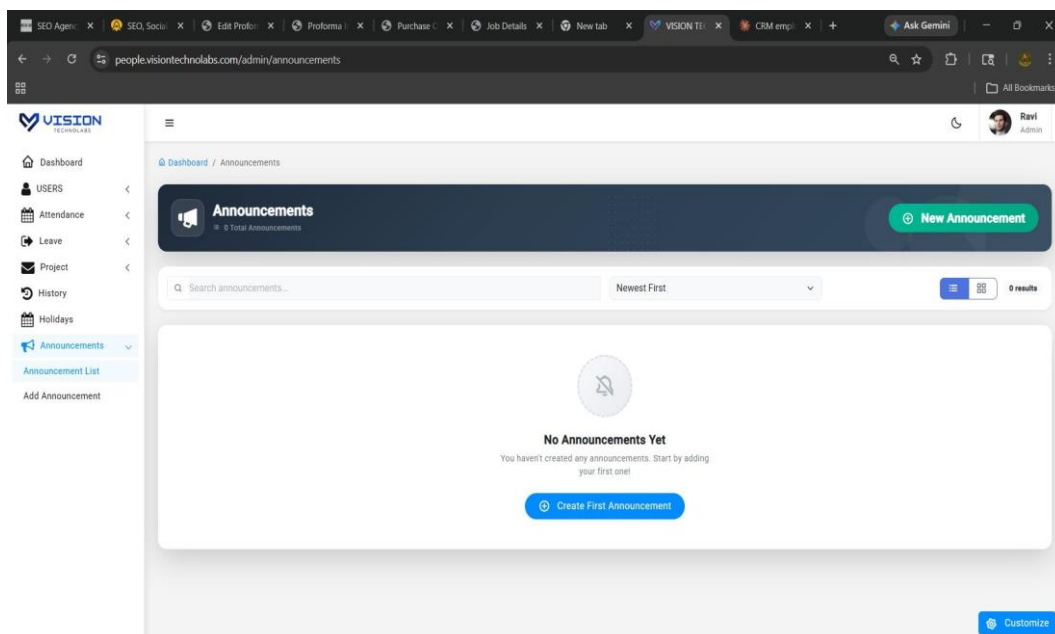


Fig 5.28 Admin – Announcements Listing

The Announcements Listing page (Fig 5.28) shows all published announcements. The dark banner header displays the total count (0 Total Announcements in this view) with a 'New Announcement' button. When no announcements exist, the page shows 'No Announcements Yet' with a 'Create First Announcement' call-to-action button. Sorting (Newest First) and search options are available.

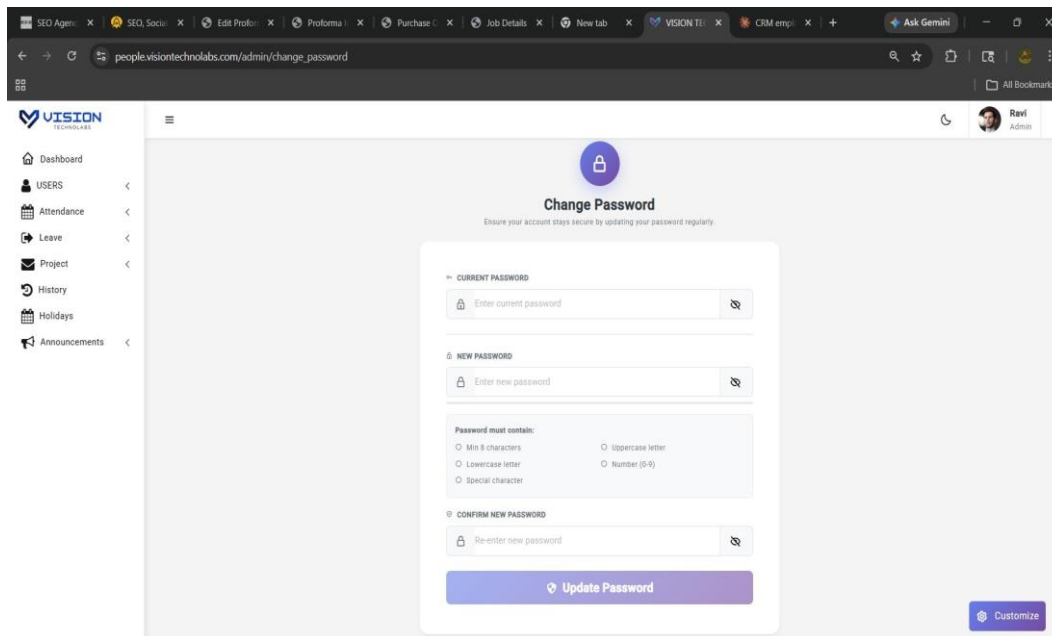


Fig 5.29 Admin – Change Password Page

The Change Password page (Fig 5.29) is available to both Admin and Employee accounts for secure password management. The form requires the Current Password, New Password, and Confirm New Password. A password strength indicator lists the requirements: minimum 8 characters, uppercase letter, lowercase letter, special character, and number (0-9). The 'Update Password' button saves the new password in BCRYPT-hashed format in the database.

5.2.3 Access Control and Security

The system implements a session-based authentication mechanism. Upon successful login, a PHP session is created storing the user's ID, role (Admin/Employee), and authentication token. All subsequent page requests are validated against this session. Direct URL access to any page without a valid session redirects the user to the login page.

Role-Based Access Control (RBAC) ensures that employees cannot access admin-only pages and vice versa. Passwords are stored in the database using PHP's `password_hash()` function with the BCRYPT algorithm. All database queries use prepared statements to prevent SQL injection attacks. The application is served over HTTPS with 256-bit SSL encryption.

CHAPTER 6 IMPLEMENTATION

6.1 IMPLEMENTATION PLATFORM / ENVIRONMENT

Table 6.1 Implementation Environment Specifications

Component	Specification
Operating System	Windows 11 (Development), Linux Ubuntu (Production)
Web Server	Apache 2.4 (XAMPP for local, Linux server in production)
Backend Language	PHP 8.1
Database	MySQL 8.0
Frontend	HTML5, CSS3, Bootstrap 5.3, JavaScript ES6+, jQuery 3.x
Code Editor	Visual Studio Code with PHP Intelephense extension
Browser Testing	Google Chrome 122, Microsoft Edge 122, Mozilla Firefox 123
Version Control	Git 2.x with GitHub repository
API Testing	Postman v10.x
Live URL	https://people.visiontechnolabs.com/

6.2 TECHNOLOGY / MODULES SPECIFICATION

6.2.1 Employee Panel Modules

- **Login Module:** Authenticates the employee using email and password. On successful authentication, a PHP session is initialized and the user is redirected to the employee dashboard.
- **Dashboard Module:** Central hub featuring the active work session timer (implemented using JavaScript setInterval), task board, break activity monitor, team directory, birthdays, announcements, and upcoming holidays.
- **Attendance Module:** Allows employees to mark daily attendance with server-side validation to prevent duplicate submissions on the same day.
- **Leave Management Module:** Multi-step form-based leave application with client-side validation. Submitted leaves are stored with 'Pending' status.

- **Task Management Module:** Displays assigned tasks with a per-task timer. Work sessions are created in the database when a task timer starts and closed with duration on stop.
- **Break Log Module:** The BREAK button triggers a break session. The employee provides a reason, and the break start time is recorded; end time and duration are auto-calculated on resume.

6.2.2 Admin Panel Modules

- **Admin Dashboard Module:** Provides high-level statistics (total employees, holidays, announcements) and embedded Manual Work Log and Manual Break Log entry forms.
- **Employee Management Module:** Three-step employee creation form (Personal, Account, Location) with profile photo upload and a searchable employee directory listing.
- **Attendance Management Module:** Real-time attendance list showing today's status for all employees, with drill-down to per-employee monthly attendance details.
- **Leave Approval Module:** Admin views all leave applications and approves or rejects them with a single click from the Leave Management page.
- **Project Management Module:** Admin creates projects and assigns tasks to employees via the Add New Task form with assignment, priority, schedule, and duration settings.
- **Work History Module:** Card-based employee selector opens detailed work and break time history for any employee for admin review and reporting.
- **Holiday Management Module:** Admin adds, edits, and deletes company holidays that are visible to all employees on their dashboard.
- **Announcement Module:** Admin publishes announcements with title and description, visible to all employees on their dashboard.
- **Password Management Module:** Secure password update with strength validation for both Admin and Employee accounts.

6.3 FINDINGS / RESULTS / OUTCOMES

The CRM Employee Management System was successfully developed and deployed on the Vision Technolabs server at people.visiontechnolabs.com. Testing with real users during the final week of the internship produced the following key outcomes:

- The attendance marking system was adopted by the team with zero training time, demonstrating the intuitiveness of the UI design.
- The active work session timer accurately tracks daily work hours with data stored for historical analysis.
- Leave applications submitted through the portal were processed within minutes, compared to hours with the previous email-based system.
- Task assignment and tracking significantly improved project visibility for the admin.
- Employee Joshi yug completed TASK-0146 (Environment Setup) in 2h 29m across 2 sessions, confirming accurate time tracking.
- The attendance list confirmed 3 employees Present and 0 Absent for April 6, 2026.
- Leave records showed 2 Approved and 1 Pending leave, confirming the end-to-end workflow.
- The admin successfully added manual work and break log entries for missed timer situations.

6.4 RESULT ANALYSIS AND DELIBERATIONS

Table 6.2 Comparison: Manual System vs. CRM EMS

Parameter	Manual System	CRM EMS (New System)
Attendance Marking	Physical register / Excel	Digital portal, real-time
Work Hour Tracking	Not available	Automatic timer-based tracking
Leave Processing	Email-based, slow	Portal-based, instant notification
Task Tracking	WhatsApp / verbal	Formal task board with time logging
Break Monitoring	Not tracked	Automated break log with duration
Manual Log Entry	Not possible	Admin can add missed entries
Project Management	No formal system	Integrated project and task module
Report Generation	Manual compilation	Automated database reports
Data Accuracy	Error-prone	High accuracy with validation

CHAPTER 7 TESTING

7.1 TESTING PLAN / STRATEGY

Testing was conducted following a structured strategy covering Unit Testing, Integration Testing, Functional Testing, User Acceptance Testing (UAT), and Browser Compatibility Testing across Chrome, Edge, and Firefox.

7.2 TEST RESULTS AND ANALYSIS

7.2.1 Test Cases

Table 7.1 Test Cases – Employee Login Module

Test ID	Test Condition	Expected Output	Actual Output	Remark
TC-01	Valid email and password	Redirect to Dashboard	Redirect to Dashboard	PASS
TC-02	Invalid email format	Show validation error	Show validation error	PASS
TC-03	Wrong password	Show error message	Error message shown	PASS
TC-04	Empty email field	Show required field error	Validation shown	PASS
TC-05	Direct URL without login	Redirect to Login	Redirect to Login	PASS

Table 7.2 Test Cases – Attendance Module

Test ID	Test Condition	Expected Output	Actual Output	Remark
TC-06	Mark attendance as Present	Saved, status: Present	Attendance saved	PASS
TC-07	Mark attendance as Absent	Saved, status: Absent	Attendance saved	PASS
TC-08	Mark attendance twice same day	Show already marked message	Error displayed	PASS
TC-09	View attendance list for month	Correct records shown	Records displayed	PASS
TC-10	Filter by different month	Filtered records shown	Filter works	PASS

Table 7.3 Test Cases – Leave Management Module

Test ID	Test Condition	Expected Output	Actual Output	Remark
TC-11	Submit leave for future date	Leave created, Pending status	Leave created	PASS

TC-12	Submit without selecting	datRequired field error	Validation error	PASS
TC-13	Submit for today date	Show advance warning	Warning displayed	PASS
TC-14	Admin approves leave	Status changes to Approved	Updated correctly	PASS
TC-15	Admin rejects leave	Status changes to Rejected	Updated correctly	PASS

Table 7.4 Test Cases – Task Management Module

Test ID	Test Condition	Expected Output	Actual Output	Remark
TC-16	Start task timer	Timer starts, session created	Timer starts	PASS
TC-17	Stop task timer	Session closed, duration logged	edDuration logged	PASS
TC-18	Mark task as Done	Task status: Completed	Task marked done	PASS
TC-19	View task detail page	Sessions and total time shown	Correct details	PASS
TC-20	Admin assigns task to em	ployeeTask in employee task board	Task visible	PASS

Table 7.5 Test Cases – Admin Functions

Test ID	Test Condition	Expected Output	Actual Output	Remark
TC-21	Add new employee	Employee in directory	Employee created	PASS
TC-22	Manual work log entry	Log in today work logs table	Log added	PASS
TC-23	Manual break log entry	Log in today break logs table	Log added	PASS
TC-24	Add new project	Project in All Projects list	Project created	PASS
TC-25	Add holiday	Holiday in company holidays	Holiday added	PASS
TC-26	Publish announcement	Announcement visible to employees	Published	PASS
TC-27	Change admin password	Password updated securely	Password changed	PASS

All 27 test cases executed during the testing phase passed successfully, with an overall pass rate of 100%. No critical bugs were found during the final testing phase. Minor UI alignment issues observed in browser compatibility testing were addressed before final deployment.

CHAPTER 8 CONCLUSION AND DISCUSSION

8.1 OVERALL ANALYSIS OF PROJECT VIABILITIES

The CRM Employee Management System developed during the internship at Vision Technolabs is a fully functional, scalable, and user-friendly web application that successfully addresses real-world workforce management challenges. The system is technically feasible, operationally viable, and economically efficient.

The dual-panel architecture covering both Employee and Admin roles provides comprehensive coverage of all daily workforce management activities. All 27 test cases passed successfully confirming functional correctness. The choice of PHP with MySQL as the backend stack proved robust and reliable.

8.2 PHOTOGRAPHS AND MENTOR VISIT DETAILS

The internship was conducted under the supervision of Internal Guide Mr. Nayan Meghwal. Regular review meetings were conducted throughout the period.

[Space reserved for photographs of work environment and mentor visit]

8.3 DATES OF CONTINUOUS EVALUATION

Evaluation	Review Type	Date
Internal Review 1	Progress Review – Requirements and Design Phase	_____
Internal Review 2	Progress Review – Implementation and Testing	_____
Final Submission	Complete Project Demonstration and Report	_____

8.4 PROBLEMS ENCOUNTERED AND POSSIBLE SOLUTIONS

Session Timer Synchronization: Keeping the JavaScript timer synchronized with server-side session data after page refreshes was challenging. Solved by storing session start time in the database and recalculating elapsed time on each page load.

Break and Work Time Calculation: Accurately calculating net work time by subtracting break durations required careful database query design with separate tracking for work and break sessions.

Multi-step Form Validation: Implementing the leave application form with proper client-side and server-side validation required careful JavaScript step navigation with checks at each transition.

Role-based Access Control: Preventing employees from accessing admin pages required PHP session checks at the top of every protected page with immediate redirect on failure.

Admin Panel – Manual Log Integration: Integrating the manual work log and break log addition into the existing session-based time tracking system required careful timestamp management to avoid data conflicts.

8.5 SUMMARY OF INTERNSHIP / PROJECT WORK

The 14-week internship at Vision Technolabs was an enriching and transformative experience. The CRM Employee Management System is a comprehensive web application covering attendance management, task tracking, leave management, break logging, project management, and organizational communication for both Employee and Admin roles.

Throughout the internship, significant skills were developed in full-stack web development (PHP/MySQL), responsive UI design (Bootstrap), JavaScript programming for real-time interactions, database design, and software testing. The professional environment enhanced time management, communication, and problem-solving skills.

8.6 LIMITATIONS AND FUTURE ENHANCEMENTS

8.6.1 Current Limitations

- Payroll and salary management is not supported in the current version.
- Mobile push notifications are not implemented.
- Advanced analytics dashboards with graphical representations are absent.
- Biometric device integration is not yet supported.
- Multi-company or multi-branch configurations are not supported.

8.6.2 Future Enhancements

- Integration of payroll management with automated salary calculation based on attendance.
- Dedicated mobile application for Android and iOS for on-the-go management.
- Advanced analytics dashboards showing attendance trends and productivity metrics.
- Biometric device integration (fingerprint/facial recognition) for automated attendance.
- Real-time chat and notification system for seamless team communication.
- Multi-company SaaS deployment for multiple organizations.
- Integration with Google Calendar/Outlook for holiday and event synchronization.
- Machine learning-based attendance pattern analysis for workforce planning.

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APPENDIX

Appendix 1 – Project Live UR

URL: <https://people.visiontechnolabs.com/>

Table A.1 Technology Stack Summary

Layer	Technology Used
Frontend	HTML5, CSS3, Bootstrap 5.3, JavaScript ES6, jQuery 3.x
Backend	PHP 8.1
Database	MySQL 8.0
Web Server	Apache (XAMPP locally, Linux server in production)
Version Control	Git 2.x / GitHub
API Testing	Postman v10.x
Code Editor	Visual Studio Code

Table A.2 Internship Details

Detail	Information
Student Name	Joshi Yug Mukeshbhai
Enrollment Number	2302020400449
Programme	Diploma Engineering in Computer Engineering
Semester	6th Semester
Institute	Aditya Silver Oak Institute of Technology
University	Silver Oak University, Ahmedabad
Internal Guide	Mr. Nayan Meghwal
Company	Vision Technolabs, Ahmedabad
Internship Duration	14 Weeks (Academic Year 2025–26)

Project Title	CRM Employee Management System
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